

1.2 The R Programming Language ↕



R is a programming language and computing environment designed for data analysis, statistics, and visualisation. It is particularly well suited to financial and time series analysis.

R is widely used in finance and economics because it:

- Is **open source and free**, lowering barriers to access
- Is supported by a **large and active research and practitioner community**
- Provides a rich ecosystem of **specialised packages** for financial data, time series modelling, and forecasting
- Enables transparent and flexible analysis through script-based workflows

Crucially, R supports **reproducibility and transparency**, which are central to credible financial analysis. Analyses can be documented, shared, replicated, and audited, allowing assumptions and results to be clearly examined. This is particularly important in financial contexts where decisions may carry significant economic consequences.

Use of R in This Course

This course uses R as the primary tool for financial analytics and time series analysis.

The course does not assume prior programming experience. Instead, it emphasises the use of established, high-level packages that allow students to focus on financial concepts, interpretation, and reasoning rather than low-level coding. The aim is not to train software developers, but to equip students with practical, transparent, and analytically sound tools for modern financial analysis.

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Why R

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Let's talk about why we are using R in this course.

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Well the first reason is reproducibility and that

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matters a lot. Picture analysis built in Excel

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with several rounds of copying and pasting

Installing R

R must be installed before it can be used.

Steps:

1. Go to the posit website: <https://posit.co/download/rstudio-desktop/> 
(<https://posit.co/download/rstudio-desktop/>).
2. Download R for your operating system (Windows, macOS, or Linux).
3. Follow the installation instructions.
4. After installation, you should be able to open the R console.

R itself is the software that performs calculations and runs code.

Installing RStudio

RStudio is not R itself. It is an integrated development environment (IDE) that makes R easier to use.

RStudio provides:

- A place to write and save code
- A console to run code
- Panels to view data, plots, and help

Install RStudio Desktop after installing R. This is available here:

<https://posit.co/download/rstudio-desktop/>  (<https://posit.co/download/rstudio-desktop/>).

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A Brief Introduction to R Studio

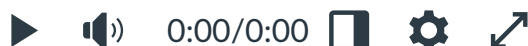


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Understanding the RStudio Interface

When you open RStudio, you will see four main panes. Understanding these panes is important before writing any code.

Source pane (top-left)

- This is where you write scripts.
- Scripts are files that contain your R code.
- Writing code in scripts allows you to save and reuse your work.

You will spend most of your time writing code here.

Console pane (bottom-left)

- This is where code is executed.
- You can type code directly here or send code from the Source pane.
- Output from calculations and messages appear here.

The Console is where R “does the work”.

Environment pane (top-right)

- This pane shows all objects currently stored in memory.
- Objects include data, variables, and results of calculations.
- Seeing objects here helps you keep track of your analysis.

If an object appears in the Environment, it means it has been created successfully.

Files, Plots, Packages, Help pane (bottom-right)

This pane changes depending on what you are doing.

- Files: browse files in your working directory
- Plots: view graphs you create
- Packages: see installed and loaded packages
- Help: view documentation for functions

You will frequently switch between Plots and Help during analysis.

R works in a simple cycle:

1. You write code in the Source pane.
2. You run the code.
3. Results appear in the Console, Plots pane, or Environment.

You can also write and run short snippets of code directly in the Console. Just type a line of code and press enter. This is great for having conversations with the data.

You do not need to understand everything at once. The goal is to become comfortable with where things appear and how to run code.